CPEN 416

Lecture 15: Order finding and Shor's algorithm

Tuesday 3 November 2026

Announcements

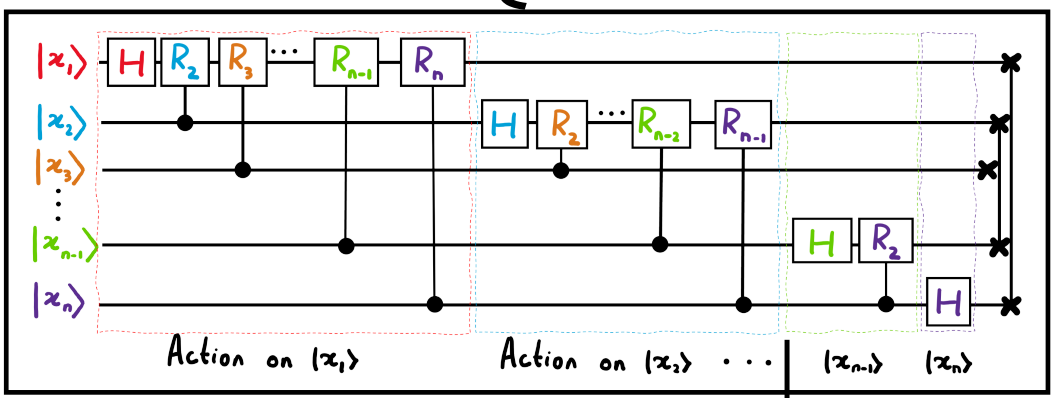
Kraus

- Quiz 7 today

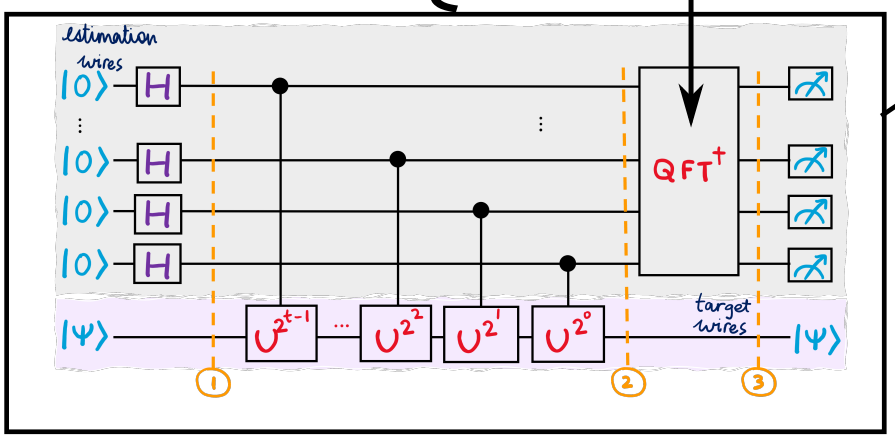
★ Projects: set up GitHub repos ASAP and do development in there

Reminder: where are we going?

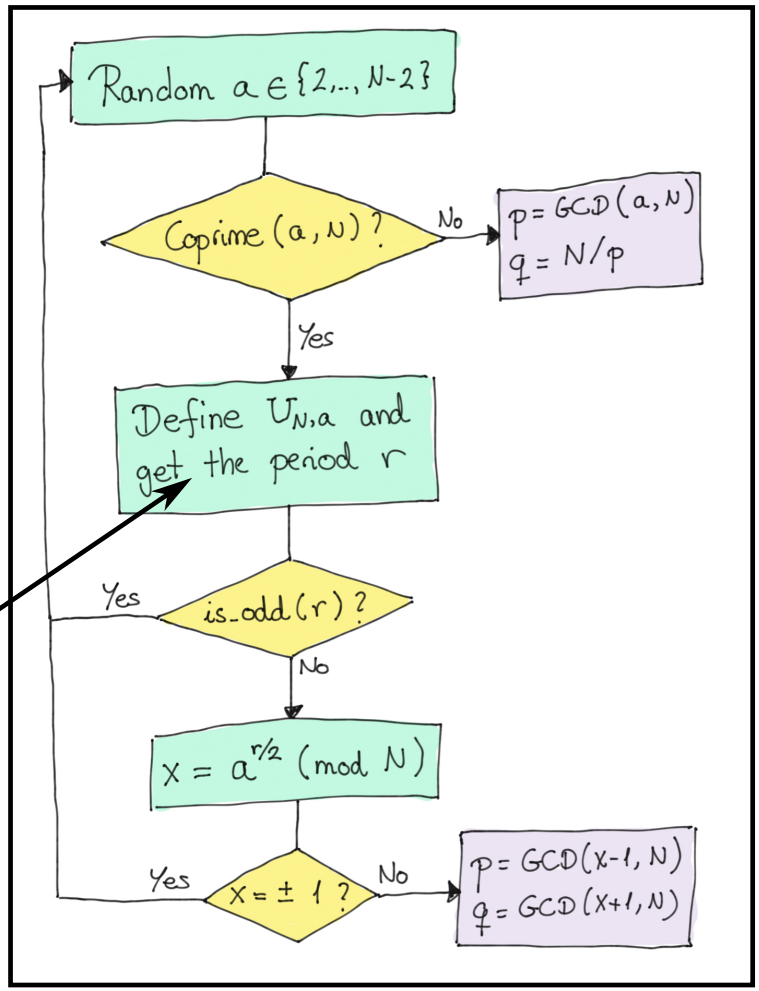
1. QFT



2. QPE



3. Shor



Last time

Input: integers a and N, where $N \geq 2$ and $\gcd(a, N) = 1$
Output: smallest integer r for which $a^r \equiv 1 \bmod N$

Exercise: find the order of $a = 8$ for $N = 15$.

$8^1 = 8$

$8^2 = 64 \equiv 4 \bmod 15$

$8^3 = 32 \equiv 2 \bmod 15$

$8^4 = 16 \equiv 1 \bmod 15$

$r = 4$

Learning outcomes

- Apply QPE to solve the order finding problem
- Reduce order finding to integer factorization, and implement Shor's quantum algorithm in PennyLane
- Explain the relevance, and the societal and ethical implications, of Shor's algorithm

Order finding on a quantum computer

$U_{7,4} |0101\rangle = |0110\rangle$

5 → 6 mod 7

Given a, N with $gcd(a, N) = 1$, consider the quantum operation for multiplication by a modulo N,

$$U_{N,a} |k\rangle = |ak \bmod N\rangle$$

$U_{15,8} |1\rangle = |8\rangle$ $(U_{15,8})^2 |1\rangle = |4\rangle$ ...

If r is the order of a modulo N, and we apply $U_{N,a}$ r times,

$$(U_{N,a})^r |k\rangle = |a^r \cdot k \bmod\rangle = |k\rangle$$

$a^r \equiv 1 \bmod N$

Eigenstate of $U_{15,8}$?? $U_{15,8} |4\rangle = |4\rangle$

$U_{15,0}$:

$$|1\rangle \mapsto |0\rangle$$
$$|0\rangle \mapsto |4\rangle$$
$$|4\rangle \mapsto |2\rangle$$
$$|2\rangle \mapsto |1\rangle$$

$$U\left(\frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)\right) \mapsto \frac{1}{\sqrt{2}}(|0\rangle + |4\rangle)$$

$$U\left(\frac{1}{\sqrt{3}}(|1\rangle + |0\rangle + |4\rangle)\right) \mapsto \cdots$$

$$U\left(\frac{1}{\sqrt{4}}(|1\rangle + |0\rangle + |4\rangle + |2\rangle)\right)$$
$$= \frac{1}{\sqrt{4}}(|0\rangle + |4\rangle + |2\rangle + |1\rangle)$$

Order finding on a quantum computer

$U_{N,a}$ has the same order, r. We can use QPE to efficiently find r with a clever initialization of the target register.

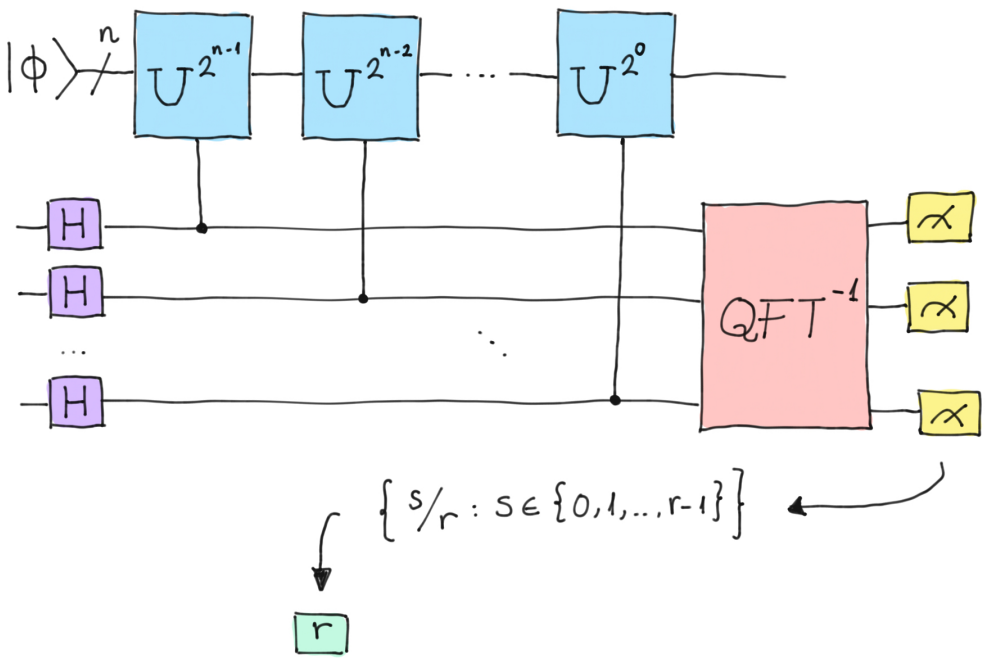


Image credit: Xanadu Quantum Codebook node S.3

Order finding on a quantum computer

Suppse there exists $|\phi\rangle$ for which

$$U_{N,a}^{r}\,|\phi\rangle = U^{r}|\phi\rangle = |\phi\rangle$$

Consider the superposition state

$$|\Psi_0\rangle = \frac{1}{\sqrt{r}}\left(|\phi\rangle + U|\phi\rangle + U^2|\phi\rangle + \cdots + U^{r-1}|\phi\rangle\right)$$

$U_{15,8}$ – $\frac{1}{\sqrt{4}}$ $|1\rangle$ $|8\rangle$ $|4\rangle$ $|2\rangle$

Apply U to $|\Psi_0\rangle$:

$$U|\Psi_0\rangle = \frac{1}{\sqrt{r}}\left(U|\phi\rangle + U^2|\phi\rangle + \cdots + U^{r}|\phi\rangle\right)$$

$$= \frac{1}{\sqrt{r}}\left(U|\phi\rangle + U^2|\phi\rangle + \cdots + |\phi\rangle\right)$$

$$= |\Psi_0\rangle$$

Order finding on a quantum computer

Now consider the state

$$|\Psi_1\rangle = \frac{1}{\sqrt{r}}\left(|\phi\rangle + e^{-\frac{2\pi i}{r}} U|\phi\rangle + e^{-2\cdot\frac{2\pi i}{r}} U^2|\phi\rangle + \cdots + e^{-(r-1)\frac{2\pi i}{r}} U^{r-1}|\phi\rangle\right)$$

Apply U to $|\Psi_1\rangle$:

$$U|\Psi_1\rangle = \frac{1}{\sqrt{r}}\left(U|\phi\rangle + e^{-\frac{2\pi i}{r}} U^2|\phi\rangle + \cdots + e^{-(r-1)\cdot\frac{2\pi i}{r}} U^r|\phi\rangle\right)$$

$$= e^{\frac{2\pi i}{r}}|\Psi_1\rangle$$

Order finding on a quantum computer

This generalizes to $|\Psi_s\rangle$

$$|\Psi_s\rangle = \frac{1}{\sqrt{r}}\left(|\phi\rangle + e^{-s\cdot\frac{2\pi i}{r}} U|\phi\rangle + e^{-2s\cdot\frac{2\pi i}{r}} U^2|\phi\rangle + \dots + e^{-(r-1)\cdot s\frac{2\pi i}{r}} U^{r-1}|\phi\rangle\right)$$

It has eigenvalue

$$U|\Psi_s\rangle = e^{\frac{2\pi i s}{r}}|\Psi_s\rangle$$

Idea: create *any* $|\Psi_s\rangle$, run QPE to get estimate of s/r, compute r.

What's the problem? $\Rightarrow$ we don't know r! we are trying to find it...

Order finding on a quantum computer

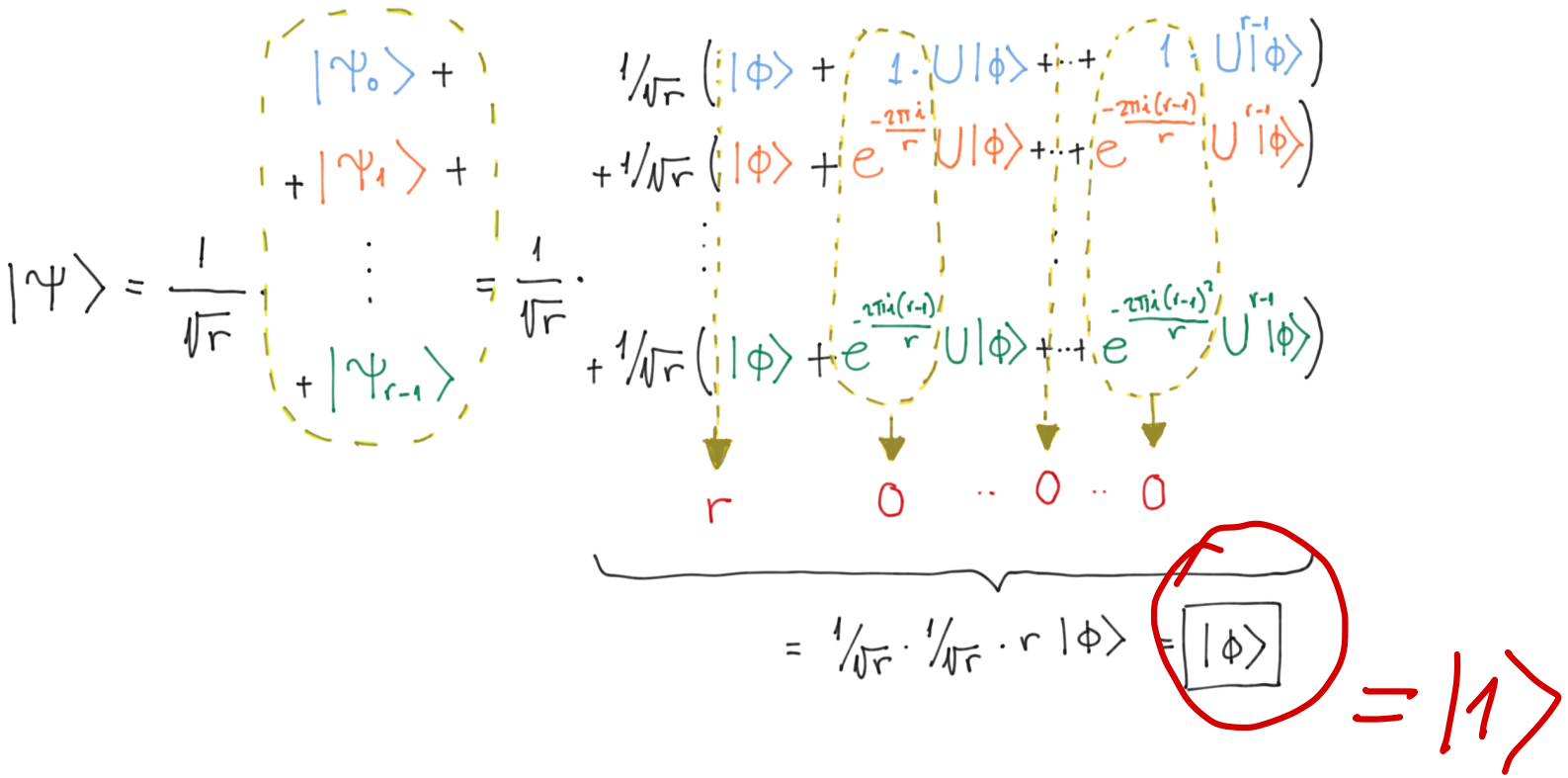
Solution: construct the uniform superposition of all of them.

$$|\psi\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\psi_s\rangle$$

But what does this equal?

Order finding on a quantum computer

The superposition of all $|\Psi_s\rangle$ is just our original state $|\phi\rangle$!



If we run QPE, the output will be s/r for one of these states.

Image credit: Xanadu Quantum Codebook node S.3

Shor's algorithm: overview

Input: integer N, guaranteed to have structure

$$N = pq$$

for p, q prime.

Output: p and q

Factoring is accomplished by a **reduction** from order finding.

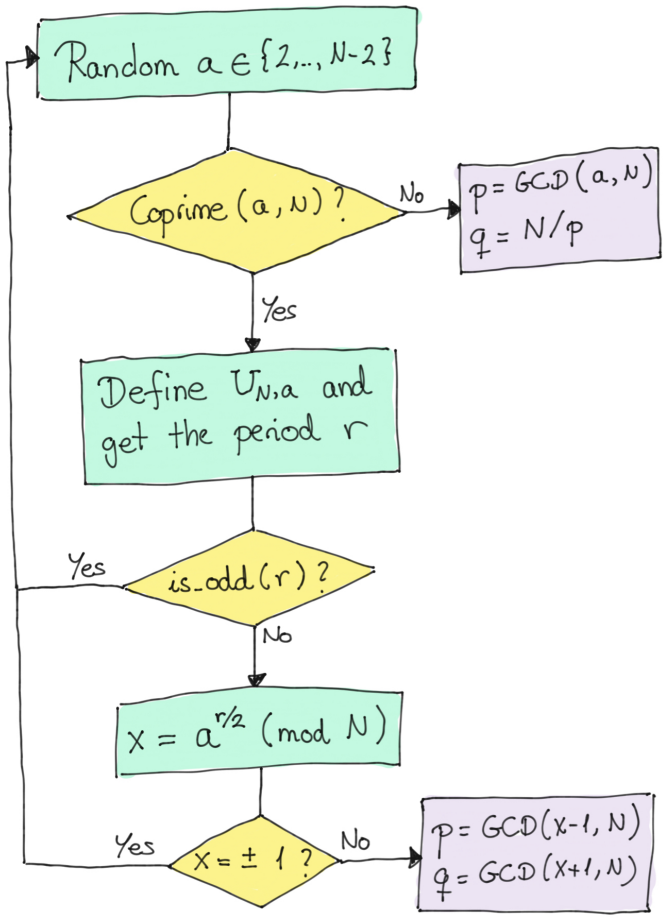


Image credit: Xanadu Quantum Codebook node S.3

Non-trivial square roots and factoring

Choose a random a that is coprime with N. If its order, r, is even,

$$a^r = a^{2r'} = (a^{r'})^2 \equiv 1 \bmod N$$

Let's rewrite and factor...

$$(a^{r'})^2 \equiv 1 \bmod N$$

$$(a^{r'})^2 - 1 \equiv 0 \bmod N$$

$$(a^{r'}+1)(a^{r'}-1) \equiv 0 \bmod N$$

Non-trivial square roots and factoring

This means

$$(a^{r'}-1)(a^{r'}+1) = kN = k \cdot p \cdot q$$

Based on structure of N,

$$a^{r'}-1 = sp$$

$$a^{r'}+1 = tq$$

If $a^{r'} \neq \pm 1$, we can determine p and q

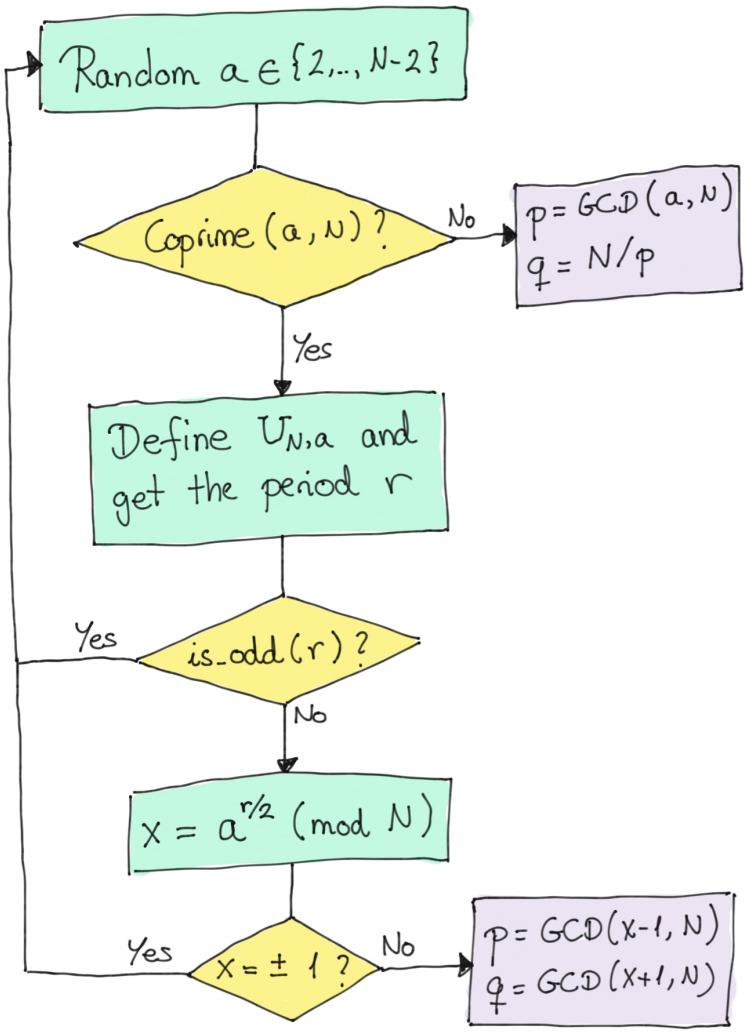
$$p = \gcd(a^{r'}-1, N)$$

$$q = \gcd(a^{r'}+1, N)$$

Is this really efficient?

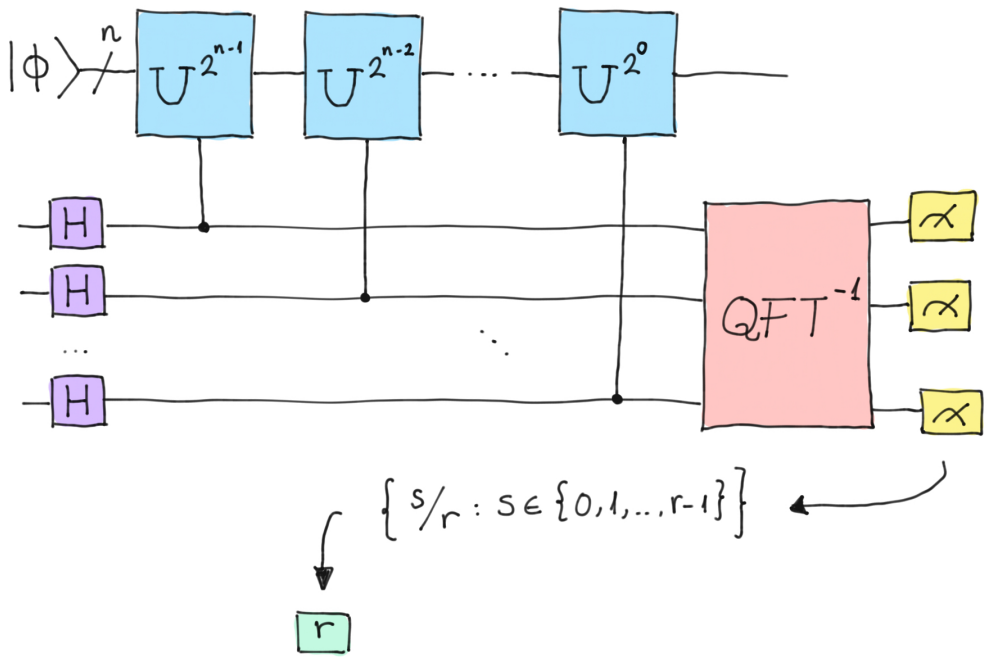
GCD: polynomial w/Euclid's algorithm

Modular exponentiation: can use exponentiation by squaring, other methods to reduce operations and memory required



Is this really efficient?

Quantum part: let $L = \lceil \log_2 N \rceil$.



QFT: gates polynomial in number of qubits $O(L^2)$

Controlled-U: requires $O(L^3)$ gates.

Discussion

Form pairs or small groups, and consider the following questions:

1. Shor's algorithm was developed in 1994. Estimate the fraction of today's world population that can actually implement it.
2. Shor's algo can be used to break cryptosystems like RSA. Estimate the proportion of the world that would be affected if someone actually deployed it at scale.
3. Is it ethical to develop such an algorithm? Is it ethical to *teach* such an algorithm?
4. Look up some resource estimates; how long would it actually take to break 2048-bit RSA? How many qubits are needed?
5. Think critically about (a) who knows how to implement the algorithm, and (b) who will potentially have access to quantum hardware in the future. What issues can you foresee?
6. What are ways we can keep our cryptographic infrastructure secure in the future?

Next time

Content:

- Starting module 4: quantum channels, and noise in quantum systems

Action items:

1. Work on project (remember to fill out surveys!)

Recommended reading:

- From this class: Codebook QFT, QPE, SH; N&C 5.3, A.5
- For next class: Codebook NT; N&C 8.1-8.3